The ORSAT decoders

Complete guide — digital modes, telegraphy, images, maritime and aeronautical messages

https://orsat.ddns.net



The FT8 decoder: audio waveform, decode markers and message table.

This guide describes, one by one, every decoder built into ORSAT: what it is for, where to listen, how to set it up and how to read what it displays. It ends with the signal identification tools and a troubleshooting chapter.

English version — October 2026 edition — ORSAT 1.0
Decoders integrated by **F1NSK** for ORSAT, based on PhantomSDR-Plus by **SV1BTL**. Signal database: **Artemis** (AresValley) and **Signal Identification Wiki**.

The screenshots show the interface in French; the button names in the text are those of the English interface.

Contents

1. Common principles 4

1.1 Where the decoders are 4

1.2 What a decoder does when it starts 4

1.3 Shared tools 5

2. FT8, FT4 and FT2 6

2.1 The three modes 6

2.2 Getting started 6

2.3 Reading the panel 6

2.4 Sync offset and Auto 7

3. JS8 (JS8Call) 8

3.1 The mode 8

3.2 Settings 8

3.3 Reading the messages 9

4. CW (Morse telegraphy) 9

4.1 How it works 9

4.2 Setting up reception 10

4.3 Regular CW stations 10

5. WSPR 10

5.1 The mode 10

5.2 Getting started 10

5.3 Reading the table 11

6. FAX (HF weather facsimile) 12

6.1 The mode 12

6.2 Choosing a station 12

6.3 Image settings 12

6.4 During reception 13

7. SSTV (slow-scan television) 14

7.1 The mode 14

7.2 Sideband 14

7.3 Settings 14

7.4 Indicators 14

8. NAVTEX 16

8.1 The mode 16

8.2 Getting started 16

8.3 The broadcast table 16

9. HFDL (HF Data Link) 17

9.1 The mode 17

9.2 Settings and reading 17

10. FLDIGI: RTTY, SITOR, PSK, Olivia and the others 18

10.1 One decoder, fourteen variants 18

10.2 Common controls 19

10.3 RTTY and SITOR (FSK) 19

10.4 PSK31 20

10.5 Olivia 20

10.6 MFSK, DominoEX, THOR, THROB, Hellschreiber, MT63 20

10.7 Packet and APRS 21

11. RADE (FreeDV digital voice) 21

12. Identification aids 21

12.1 Digital signal analysis 21

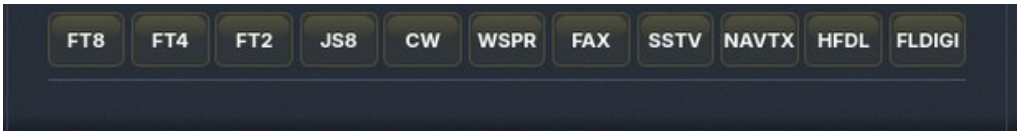
12.2 Identify a signal (Artemis database) 22

13. Troubleshooting 23

1. Common principles

1.1 Where the decoders are

All the decoders are grouped in the **Decoders Control** row, in the right-hand column, below the scanner. Each button starts a decoder; a second click on the same button stops it. **Only one decoder runs at a time**: starting another one stops the previous one.



The “Decoders Control” row and its eleven buttons.

Button	Decoder	Chapter
FT8 / FT4 / FT2	WSJT weak-signal modes	2
JS8	JS8Call	3
CW	Morse telegraphy	4
WSPR	WSPR-2 beacons	5
FAX	HF weather facsimile (WEFAX)	6
SSTV	Slow-scan television	7
NAVTX	NAVTEX (navigational warnings)	8
HFDL	HF Data Link (aviation)	9
FLDIGI	fldigi family: RTTY, SITOR, PSK31, Olivia, MFSK…	10

The **RADE (FreeDV)** digital voice modes are not in this row: they have their own **RADEL / RADEU** buttons in the mode selector (chapter 11).

1.2 What a decoder does when it starts

A decoder does more than listen to the audio: it **takes control of the receiver** to put itself in the right conditions.

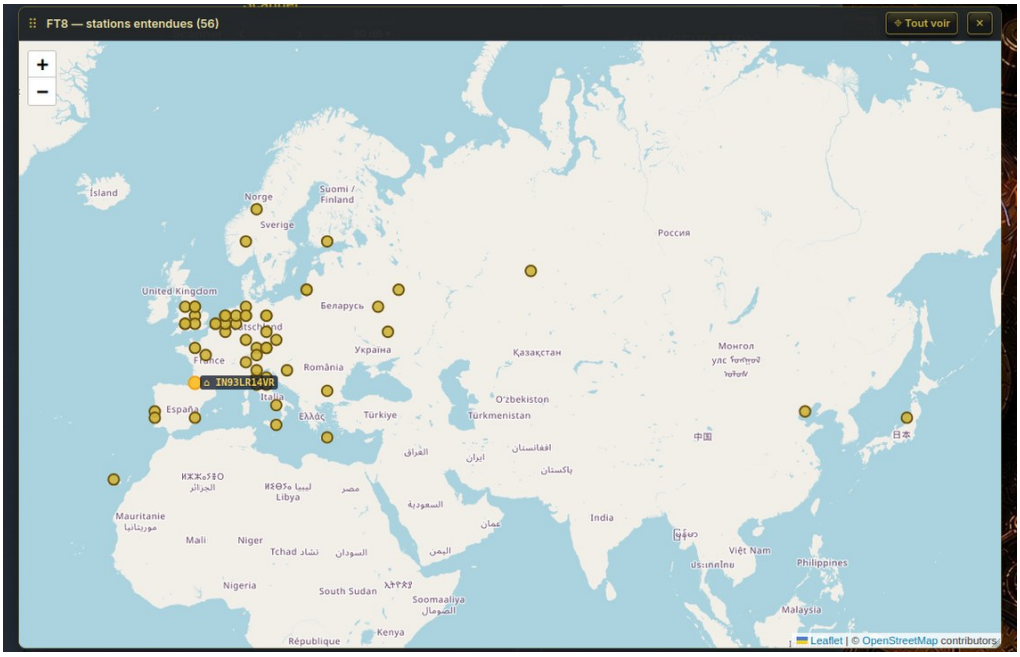
- **Demodulation mode** — FT8, FT4, FT2, JS8 and WSPR switch the receiver to **USB**, whatever the band's usual mode. FAX and NAVTEX also work in USB. SSTV picks the customary sideband (see chapter 7).
- **IF filter** — the passband is set to the mode's: the whole 3 kHz sub-band for FT8/FT4/FT2/JS8, the narrow beacon window for WSPR.
- **Back to normal** — when the decoder stops, the receiver returns to the **band's default mode** (LSB, CW, AM…).
- **Automatic analysis** — if the “Digital signal analysis” module was running, it stops: it has done its job, which was to help choose a decoder.

> ***Tip :*** *The decoders work on the receiver's **raw** audio, before noise reduction, audio AGC and volume. Muting the sound or lowering the volume therefore does not stop decoding.*

1.3 Shared tools

Known frequencies. The FT8/FT4, JS8, WSPR and SSTV panels have a **Known frequencies** menu: choosing a band tunes the receiver straight to the mode's usual dial frequency. FAX, NAVTEX, HFDL and FLDIGI have their own station lists, described in their chapters.

Station map. FT8/FT4/FT2, JS8 and WSPR have a **Map (n)** button: every station whose message contains a locator is placed on it. **Large view** opens the map in a floating window that can be moved and resized. Hovering over a dot shows the callsign and the message; a click opens the station's QRZ page.



Map of the stations heard, in large view.

Decoder audio spectrum. FT8/FT4/FT2, JS8 and WSPR show, below their settings, an **audio waterfall** of the passband: signals can be seen arriving, and each decode is marked on it.

Time synchronisation. Fixed-slot modes (FT8, FT4, FT2, JS8, WSPR) depend on UTC time. ORSAT compensates for the delay of the audio stream between the server and your browser with the **Sync offset** setting (section 2.4).

2. FT8, FT4 and FT2

2.1 The three modes

These three WSJT-X modes carry short standardised messages (callsigns, locator, report) with very high sensitivity. They work in **slots** synchronised to UTC.

Mode	Slot length	Use
FT8	15 s	The most widespread: DX, propagation, everyday operating
FT4	7.5 s	Faster, mainly in contests
FT2	3.75 s	Very fast variant, still little used

2.2 Getting started

1. Click **FT8**, **FT4** or **FT2**.
2. In **Known frequencies**, choose a band (FT8 and FT4; FT2 has no list yet). The receiver tunes in USB.
3. Wait for the current slot to end: the first decodes appear after 15 s (FT8), 7.5 s (FT4) or 3.75 s (FT2).

Band	FT8 (kHz)	FT4 (kHz)
160 m	1,840.0	—
80 m	3,573.0	3,575.0
60 m	5,357.0	—
40 m	7,074.0	7,047.5
30 m	10,136.0	10,140.0
20 m	14,074.0	14,080.0
17 m	18,100.0	18,104.0
15 m	21,074.0	21,140.0
12 m	24,915.0	24,919.0
10 m	28,074.0	28,180.0
6 m	50,313.0	50,318.0

2.3 Reading the panel



FT8 panel: waveform, sync setting and message table.

- **Farthest** — distance of the farthest station heard since the start of the session.
- **Audio waveform** — the markers show each decoded signal within the passband.
- **Message table** — one row per decode:

Column	Content
Mode	FT8, FT4 or FT2
dB	Estimated signal-to-noise ratio (a signal at −20 dB still decodes)
Hz	Audio frequency of the signal within the passband
DT	Time offset, in seconds
Message	The text: CQ F1ABC JN03, F1ABC DL1XYZ -12, RR73…
Locator / Dist	The station's Maidenhead locator and estimated distance from ORSAT

2.4 Sync offset and Auto

The audio takes a little time to reach you. **Sync offset** delays the start of the capture after the start of the UTC slot to compensate for that delay (value shown in seconds).

- **Auto** ticked (recommended) — ORSAT adjusts the offset itself from the **DT** values measured by the decoder.
- **Auto** unticked — set the slider by hand: if most DT values are strongly positive or negative, move it the same way.

> ***Tip :*** *DT values all close to 0 (between −0.5 and +0.5 s) mean the timing is right. DT values scattered beyond ±1 s lose decodes.*

3. JS8 (JS8Call)

3.1 The mode

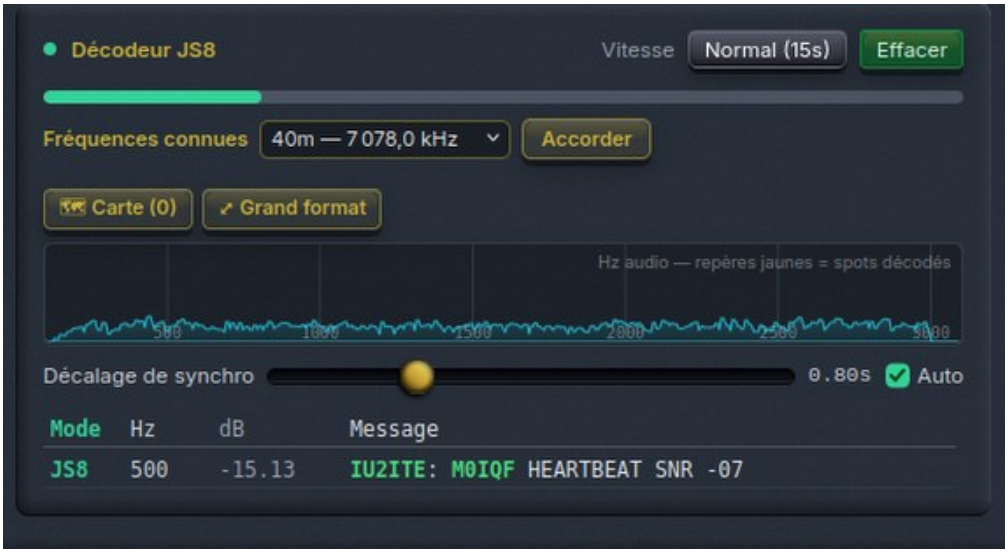
JS8 uses FT8's modulation but carries **free text**: keyboard messaging, relaying between stations, presence beacons. A long message is split into several frames.

3.2 Settings

- **Speed** — the JS8 speed being received. All stations in a QSO use the same one; **Normal** is the usual calling speed.

Speed	Slot
Normal	15 s
Fast	10 s
Turbo	6 s
Slow	30 s
Ultra	4 s

- **Known frequencies** — 160 m 1,842.0 · 80 m 3,578.0 · 40 m 7,078.0 · 30 m 10,130.0 · 20 m 14,078.0 · 17 m 18,104.0 · 15 m 21,078.0 · 12 m 24,922.0 · 10 m 28,078.0 kHz.
- **Progress bar** — progress of the current slot.
- **Sync offset / Auto** — as for FT8 (section 2.4).
- **Clear** — clears the messages.



JS8 panel: speed, slot progress and messages.

3.3 Reading the messages

Each row shows the mode, the SNR and the message, with the **sender's callsign highlighted**. A message being received is shown **as it arrives**, followed by a ▌ cursor, before being completed by the next frames. If the last frame does not arrive, the row says so (timed out).

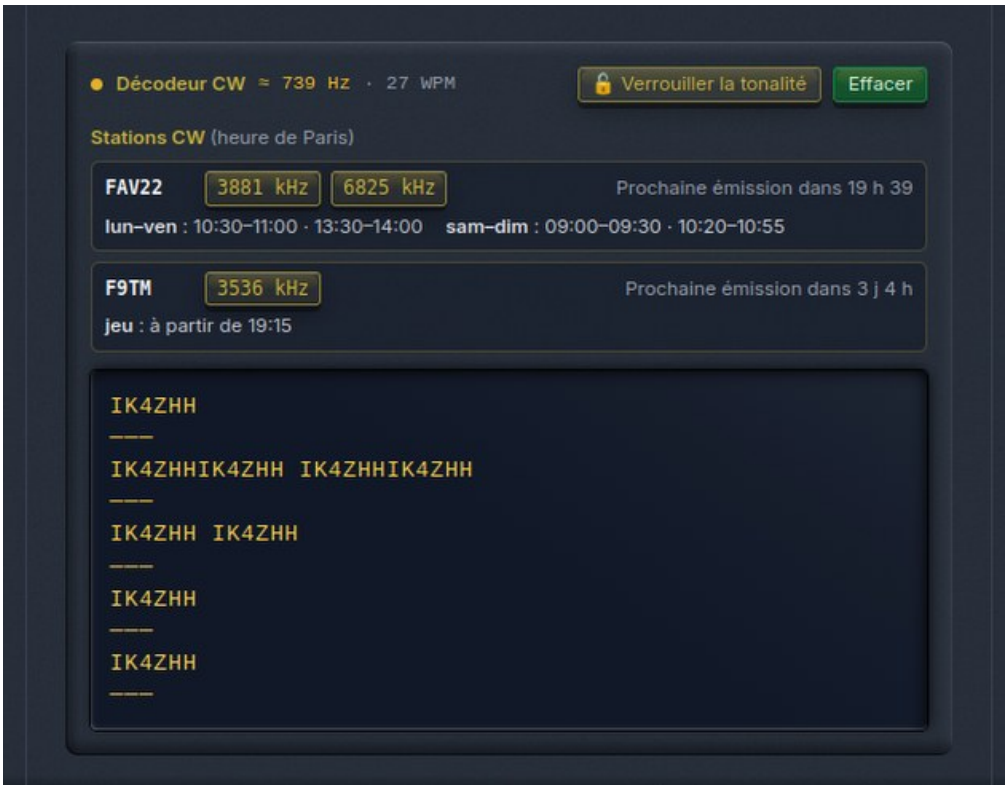
> ***Tip :*** *“Listening on the Normal cycle…” means the decoder is waiting. On a quiet band it can take several minutes before the first message.*

4. CW (Morse telegraphy)

4.1 How it works

The CW decoder **finds the signal's tone** in the passband on its own and **measures the speed** on its own. It follows speeds of roughly **5 to 65 words per minute**.

- **≈ … Hz** — audio frequency of the note being decoded.
- **· … WPM** — measured speed.
- **scanning…** — no note has been caught yet.
- **Lock tone** — holds the decoder on the current note: it no longer goes looking for a nearby signal. A second click releases it.
- The text is written continuously; a —— line marks a silence (end of transmission).
- **Clear** clears the text.



CW panel: detected note, speed and decoded text.

4.2 Setting up reception

1 Listen in **CW** mode with a narrow filter (300 to 500 Hz).
2 Put **a single signal** in the passband: two notes at once make the decoder lose sync.
3 Wait a few characters while the speed is measured.

4.3 Regular CW stations

Below the decoded text, a table lists the **scheduled CW transmissions** (Paris time, daylight-saving change handled), with a button to tune straight to them:

Station	Frequencies	Schedule
FAV22 (French Navy)	3,881 and 6,825 kHz	weekdays 10:30–11:00 and 13:30–14:00; weekends 09:00–09:30 and 10:20–10:55
F9TM	3,536 kHz	Thursday from 19:15

Tip : *FAV22's transmissions are ideal for practice: steady speed and well-formed text.*

5. WSPR

5.1 The mode

WSPR (“whisper”) is a mode for **very low-power beacons** used to study propagation. Each transmission lasts almost 2 minutes and starts at the beginning of an **even UTC minute**. ORSAT's decoder receives classic WSPR-2.

5.2 Getting started

1 Click **WSPR**.
2 Choose a band in **Known frequencies**:

Band	kHz	Band	kHz
2200 m	136.0	30 m	10,138.7
630 m	474.2	20 m	14,095.6
160 m	1,836.6	17 m	18,104.6
80 m	3,568.6	15 m	21,094.6
80 m EU	3,592.6	12 m	24,924.6
60 m	5,287.2	10 m	28,124.6
40 m	7,038.6		

1 Wait: “Waiting for next even UTC minute slot…” is shown until the next slot starts, then the progress bar moves during reception. The results arrive **at the end** of the slot.

5.3 Reading the table



WSPR panel: slot progress and decoded beacons.

- **UTC** — time of the slot;
- **Callsign** and **Grid** — the beacon's callsign and locator;
- **Power** — declared power (in dBm: 37 dBm = 5 W, 30 dBm = 1 W, 23 dBm = 200 mW);
- **Freq** — exact frequency of the beacon;
- **SNR** — signal-to-noise ratio (WSPR decodes down to about −28 dB).

> ***Tip :*** *The **WSPR log** and **WSPR map** buttons in the “External links” section of the Information menu show the beacons received by ORSAT on the reference websites.*

6. FAX (HF weather facsimile)

6.1 The mode

Meteorological and military stations broadcast **weather charts** (analyses, forecasts, sea state, ice) by radio facsimile. Each image is preceded by a start and phasing signal and ended by a stop signal.

6.2 Choosing a station

- **Station** — choose a station from the list, then a **Frequency**, and click ▶ **Tune**.

Station	Frequencies (kHz)
GYA — Northwood (United Kingdom)	2,618.5 · 4,610 · 8,040 · 11,086.5
DDH3 / DDK6 — Hamburg (Germany)	3,855 · 7,880 · 13,882.5
SVJ4 — Athens (Greece)	4,482.9 · 8,106.9
RBW41 — Murmansk (Russia)	5,336 · 6,445.5 · 7,908.8 · 8,444 · 10,130
HLL2 — Seoul (South Korea)	3,585 · 5,857.5 · 7,433.5 · 9,165 · 13,570
JMH — Tokyo (Japan)	3,622.5 · 7,795 · 13,988.5
JFX — Kagoshima (Japan)	4,274 · 8,658 · 13,074 · 16,907.5 · 22,559.6

- **Next Transmissions · UTC** — the chosen station's next transmissions, with a countdown. A transmission **in progress** is shown in green (●), an **imminent** one in red. The **FAX transmission schedules** link opens the full schedules.

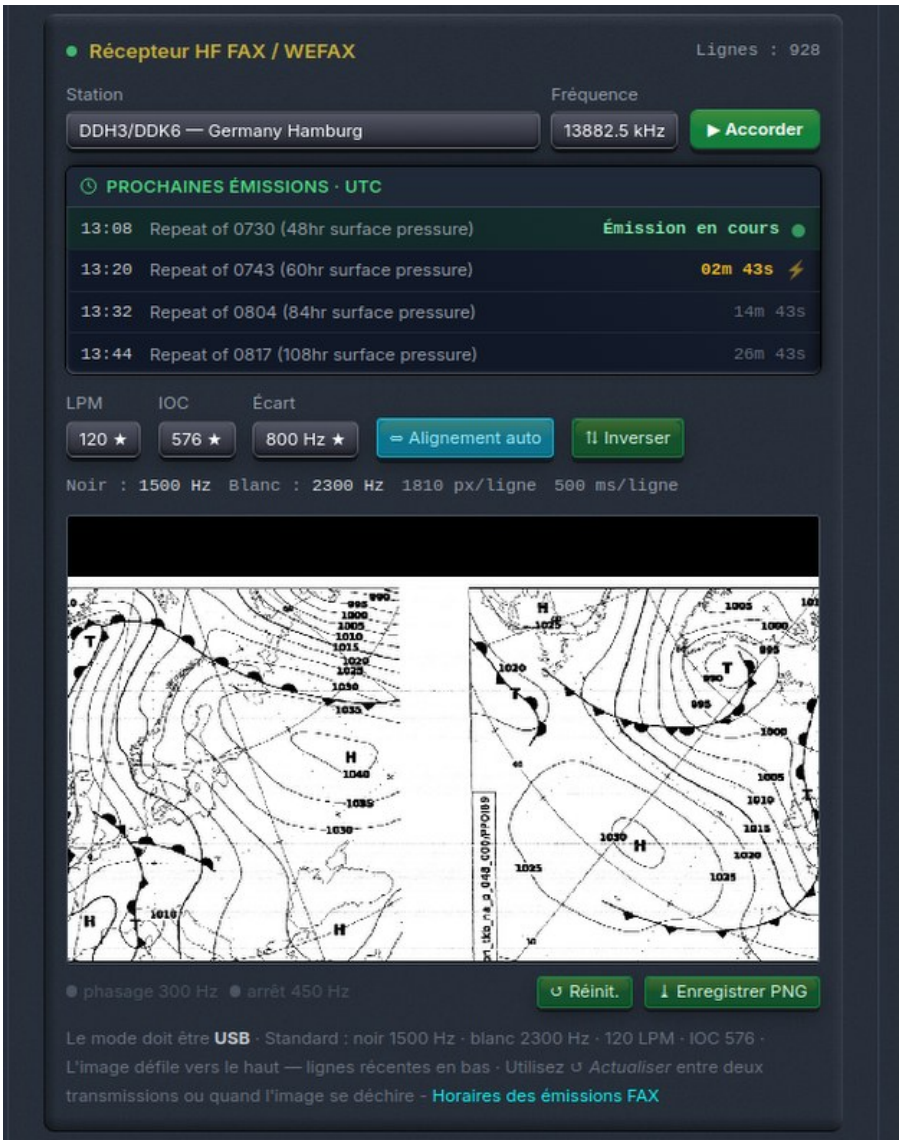
6.3 Image settings

Setting	Values	Default
LPM (lines per minute)	100 · 120 · 240	120
IOC (index of cooperation)	288 · 576	576
Shift (black/white)	400 Hz · 800 Hz	800 Hz

The values marked ★ in the menus are the standard; almost every station uses it. The information line shows the **black** (1,500 Hz) and **white** (2,300 Hz) frequencies, the resolution in pixels per line and the duration of a line.

- ⇔ **Auto-align** — automatically aligns the lines on the sync pulses: the image no longer slants.
- ⇅ **Invert** — swaps black and white (LSB reception or a station with reversed polarity).

6.4 During reception



FAX panel: station, schedule, settings and image being received.

- The **300 Hz phasing** and **450 Hz stop** lights come on during phasing and the stop signal; the header shows **[PHASING]**, **[STOP]** and the number of lines received (**Lines**).
- The image builds line by line, **newest lines at the bottom**.
- **↺ Reset** clears the image and resets the decoder: use it between two transmissions or if the image tears.
- **Save PNG** saves the image.

> ***Tip :*** *Start the decoder a minute or two before the scheduled time: you will catch the phasing and the image will start properly aligned.*

7. SSTV (slow-scan television)

7.1 The mode

Radio amateurs exchange **still images** in SSTV; the International Space Station also broadcasts them during special events. An image takes from 36 seconds to several minutes depending on the mode.

7.2 Sideband

By convention, SSTV is worked in **LSB on 80 and 40 m** and in **USB from 30 m up**. On start-up, the decoder picks the matching sideband. If a station does not follow the convention, change the sideband by hand: ORSAT respects your choice. On the wrong sideband, nothing decodes.

7.3 Settings

- **Known frequencies** — 80 m 3,730.0 (Europe) and 3,845.0 (Americas) · 40 m 7,171.0 · 20 m 14,230.0 (main) and 14,233.0 · 17 m 18,160.0 · 15 m 21,340.0 · 12 m 24,975.0 · 10 m 28,680.0 kHz.
- **Mode** — **Auto** (recommended) recognises the mode from the VIS header sent at the start of each image. A mode can also be forced: Martin M1, Martin M2, Scottie S1, Scottie S2, Scottie DX, Robot 36, Robot 72.
- ▶ **Start** / ■ **Stop** — starts or stops reception.
- ↺ **Reset** — clears the image.
- **Force** — starts decoding at once in the chosen mode, without waiting for the VIS header. Useful when joining in the middle of an image. Does not work in Auto mode.
- **Save** — saves the image.

7.4 Indicators



SSTV panel: detected mode, sync status and received image.

- **[detected mode]** — for example [Martin M1].
- **hard sync lock** — firm sync on the line pulses; **soft sync hold** — sync held despite a weak signal.
- **Running / Stopped** and **Lines** — status and number of lines received out of the image total.

> ***Tip :*** *A slanted image comes from poor sync; an image with wrong colours, from the wrong mode or the wrong sideband.*

8. NAVTEX

8.1 The mode

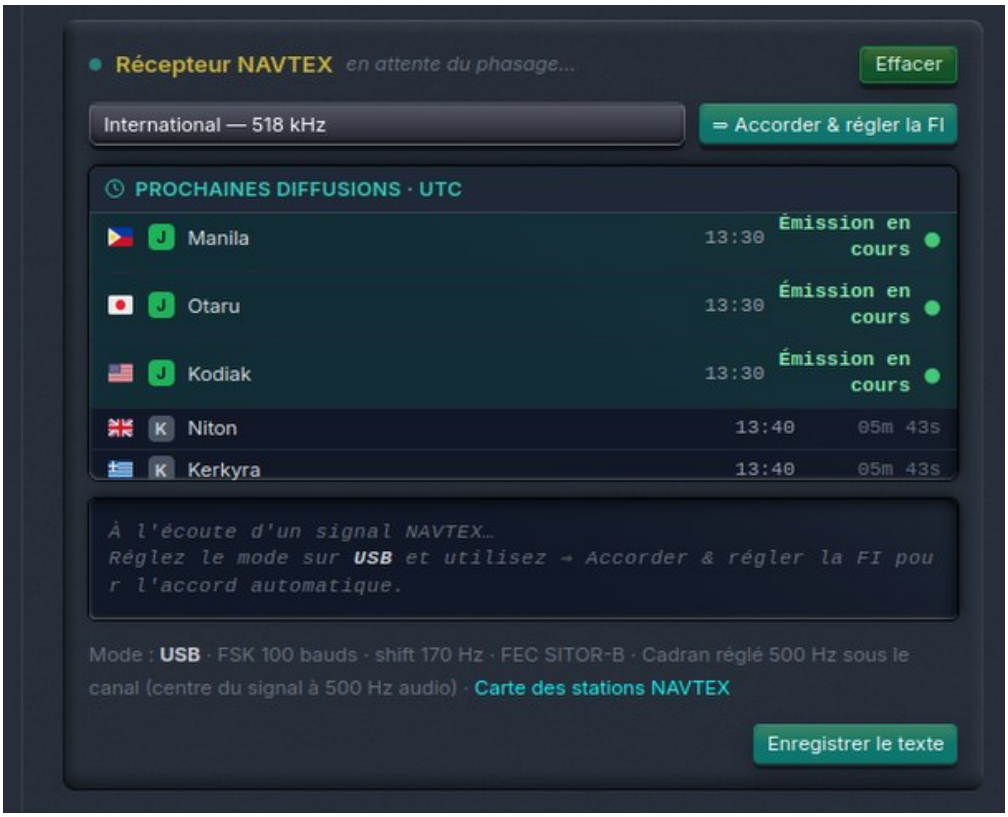
NAVTEX broadcasts **navigational warnings**, marine weather and distress alerts to ships. Each coast station has an **identification letter** (A to Z) and 10-minute transmission slots repeated every 4 hours. Transmission: 100-baud FSK, 170 Hz shift, SITOR-B code.

- **518 kHz** — international NAVTEX, in English;
- **490 kHz** — national NAVTEX, in the country's language.

8.2 Getting started

1. Click **NAVTX**: the receiver switches to USB.
2. Choose a station and click ⇒ **Tune & Set IF**: the receiver tunes 500 Hz below the channel (the signal falls at 500 Hz audio) and the IF filter is set.
3. The header shows “waiting for phasing…” until a transmission starts.

8.3 The broadcast table



NAVTEX panel: next broadcasts and messages received.

Next Broadcasts · UTC lists the stations in the order of their next transmissions: flag, identification letter, name, area, UTC time and countdown. The station **currently transmitting** is shown in green, the next one in red when it is imminent.

- A button **arms the decoder immediately**, without waiting for the phasing.
- **NAVTEX Maps Stations** opens the station map.
- **Clear** and **Save Text** clear or save the received text.

> ***Tip :*** *At night, 518 kHz carries far: stations from all over Western Europe and the Mediterranean are regularly received.*

9. HFDL (HF Data Link)

9.1 The mode

Aircraft flying over the oceans exchange **ACARS** messages with HF ground stations around the world. ORSAT's decoder is **dumphfdl**, compiled to run in your browser.

9.2 Settings and reading



HFDL receiver: ground station, indicators and messages.

- **HFDL ground station** — Shannon, Reykjavik, Canarias, Riverhead, Johannesburg… then one of its frequencies and tune to it, which also sets the IF.
- **Dial** — shows whether you are off an HFDL frequency.
- **Noise**, **Frames**, **Rate** — reception quality.
- **Messages** and **Squitters** — counters; tick **Squitters** to show the ground station beacons as well.
- Hover over a message to see the full frame; the save and clear buttons manage the log.

Clickable identifiers. The aircraft's **flight number**, **registration** and **ICAO code** open its FlightAware page in a new tab.

Aircraft map. Map shows the aircraft whose position has been received; **Large view** opens it in a floating window; **Show all** zooms to all the aircraft. Each aircraft is drawn pointing along its heading, with its track in red; a click opens its FlightAware page.

> ***Tip :*** *Ground stations send a squitter every 32 s. Choose a frequency where propagation is open towards the station and wait a few minutes: only some messages carry a position.*

10. FLDIGI: RTTY, SITOR, PSK, Olivia and the others

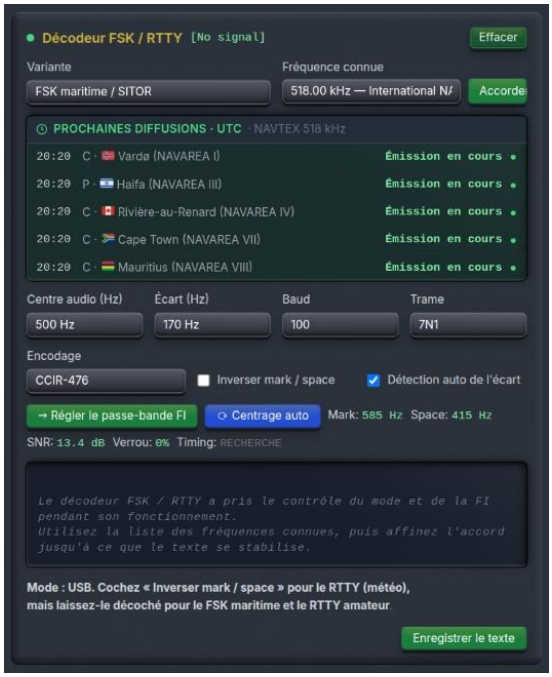
10.1 One decoder, fourteen variants

The **FLDIGI** button opens a single decoder that covers the modes of the fldigi software. The **Variant** menu chooses the family; the panel title changes accordingly (FSK / RTTY, PSK31, Olivia, Packet, APRS…).

Variant	Use
Maritime FSK / SITOR	NAVTEX, SITOR maritime warnings, NBDP
Weather RTTY	Weather bulletins in RTTY (DWD)
Amateur RTTY	Amateur radio RTTY
PSK31 (BPSK)	Amateur keyboard conversation
Olivia	Very robust text in difficult conditions
MFSK16 · 32 · 64	Robust text, copes well with fading
DominoEX	Text, resistant to drift and multipath
THOR	DominoEX with error correction
THROB / THROBX	Slow multi-tone mode
Hellschreiber	Text sent as an image (character facsimile)
MT63	Wide, very robust mode, often used for emergencies
Packet (AX.25)	Packet radio traffic
APRS	APRS positions and messages

Tip : *While running, this decoder takes control of the mode and the IF filter. First choose a known frequency, then fine-tune until the text settles.*

10.2 Common controls



FLDIGI decoder, maritime FSK / SITOR variant.

- **Known frequency** then **Tune** — tunes to a reference frequency for the variant.
- **Next Broadcasts · UTC** — broadcast times for the variants that have them (NAVTEX, DWD stations).
- ⇒ **Set IF Band-Pass** — sets the IF filter to the width of the mode.
- ⟳ **Auto-tune Center** — finds the signal and centres the decoder on it.
- **Clear** — clears the text; a button **saves the text**.
- The header shows the decoder status; “waiting for lock…” as long as no signal is caught.

10.3 RTTY and SITOR (FSK)

The three FSK variants share the same settings:

- **Center audio (Hz)** — audio frequency of the centre of the signal.
- **Shift (Hz)** — spacing between the mark and space tones.
- **Baud** — speed.
- **Framing** — 5N1, 5N1.5, 5N2, 7N1, 7E1, 7O1, 8N1.
- **Encoding** — CCIR-476 (SITOR), ITA2 / Baudot (RTTY), ASCII.
- **Invert mark / space** — swaps the two tones.
- **Auto shift detect** — measures the shift automatically.
- **Squelch (SNR)** — stops noise from being printed. Noise alone reads about −12 to −15 dB; RTTY stays readable down to about −4 dB. Slider fully left: squelch off (“off”).

Variant	Centre	Shift	Baud	Code	Invert
Maritime FSK / SITOR	500 Hz	170 Hz	100	CCIR-476, 7N1	no
Weather RTTY	1,000 Hz	450 Hz	50	ITA2, 5N1.5	**yes**
Amateur RTTY	1,000 Hz	170 Hz	45.45	ITA2, 5N1.5	no

Indicators: **Mark** and **Space** (measured frequencies of the two tones), **Tones**, **SNR**, lock and timing.

Known frequencies:

- *Maritime* — 518 kHz (international NAVTEX), 490 kHz (national NAVTEX), 2,174.5 kHz (NBDP), 4,209.5 kHz (HF NAVTEX), SITOR MSI on 4,210, 6,314, 8,416.5, 12,579, 16,806.5, 19,680.5, 22,376 and 26,100.5 kHz.
- *Weather* — DWD Pinneberg: 4,583 (DDK2), 7,646 (DDH7), 10,100.8 (DDK9), 11,039 (DDH9), 14,467.3 (DDH8) and 147.3 kHz (DDH47, longwave, 85 Hz shift).

- *Amateur* — RTTY on 3,590, 7,043, 10,143, 14,083, 21,083 and 28,083 kHz.

> ***Tip :*** *DWD weather RTTY arrives with the mark tone at the bottom: leave* ***Invert mark / space*** *ticked. For amateur RTTY and SITOR, leave it unticked.*

10.4 PSK31

- **Center audio** — put the decoder on the signal's trace (a thin line 31 Hz wide on the waterfall), or use ⟳ **Auto-tune Center**.
- Indicators: **Carrier** (carrier frequency), **IMD** (transmission quality: −25 dB or below is clean, above −20 dB the transmitter is overdriven), **DCD** (● when a signal is detected), **Audio** (level).
- Known frequencies: 3,580.15 · 7,040.15 · 10,142.15 · 14,070.15 · 18,100.15 · 21,080.15 · 24,920.15 · 28,120.15 kHz.

10.5 Olivia

- **Mode** — number of tones / width: 8/250, 16/500, 32/1000, 16/1000. It must match the station being received.
- **Squelch (FEC S/N)** — below 3.5 noise starts printing; a good signal reads 8 to 9.
- Known frequencies: 20 m 14,072.5 (8/250) and 14,108.5 (32/1000), 80 m 3,583, 40 m 7,040 and 7,072.5, 12 m 24,922.5, 10 m 28,122.5 kHz…

10.6 MFSK, DominoEX, THOR, THROB, Hellschreiber, MT63

These modes all have two settings: a **Mode** (sub-mode, chosen to match the station being received) and a **Squelch** specific to each family. The **Centre**, **Mode** and **Audio** indicators follow reception.

Variant	Sub-modes	Default
MFSK	MFSK16, MFSK32, MFSK64	MFSK16
DominoEX	Micro, 4, 5, 8, 11, 16, 22, 44, 88	DominoEX 11
THOR	Micro, 4, 5, 8, 11, 16, 22, 25 x4, 50 x1, 50 x2, 100	THOR 16
THROB	THROB 1, 2, 4 · THROBX 1, 2, 4	THROB 2
Hellschreiber	Feld Hell, Slow Hell, Hell X5, Hell X9, FSK Hell 245, FSK Hell 105, Hell 80	Feld Hell
MT63	500, 1000, 2000 (short or long interleave)	MT63-1000 long

> ***Tip :*** *In Hellschreiber the text is not decoded but* ***drawn****: you read it by eye, like a tape. Double-height text is normal.*

10.7 Packet and APRS

- **Speed** — 1,200 baud (VHF) or 300 baud (HF); 1,200 / 2,200 Hz tones on VHF.
- **Show raw packet** — also shows the raw AX.25 frame.
- Indicators: **Frames** (valid frames received), **Stations** (number of different stations), **DCD**, **Audio**.
- In APRS, stations sending a position can be **shown on the map**.
- Known frequencies: 144.800 MHz (APRS Europe), 144.390 (North America), 145.175 (Australia), 144.640 (China), 144.660 (Japan), 145.825 MHz (ISS digipeater) and 10,147.6 kHz (HF APRS, 300 baud).

> ***Tip :*** *The VHF frequencies can only be used if ORSAT's receiver covers the 2 m band.*

11. RADE (FreeDV digital voice)

RADE v1 is FreeDV's neural-network digital voice mode. It is not started from the decoders row but from the **RADEL** (LSB, for 160, 80 and 40 m) and **RADEU** (USB, for 20, 17, 15 and 10 m) buttons of the mode selector. Decoding is done **on the server**.

- **Green** light: synced, with the SNR; **yellow**: searching for a signal; **red**: the server's decoding service cannot be reached.
- Bandwidth: 1,500 Hz; the voice is rebuilt by the FARGAN vocoder.
- **FreeDV Reporter** shows the list of FreeDV stations active worldwide (qso.freedv.org).

12. Identification aids

12.1 Digital signal analysis

The module, at the head of the decoders row, **tries to recognise the mode** of the signal being received (FT8, WSPR, RTTY, PSK, Olivia, CW…) and suggests the matching decoder. This feature is **experimental**.

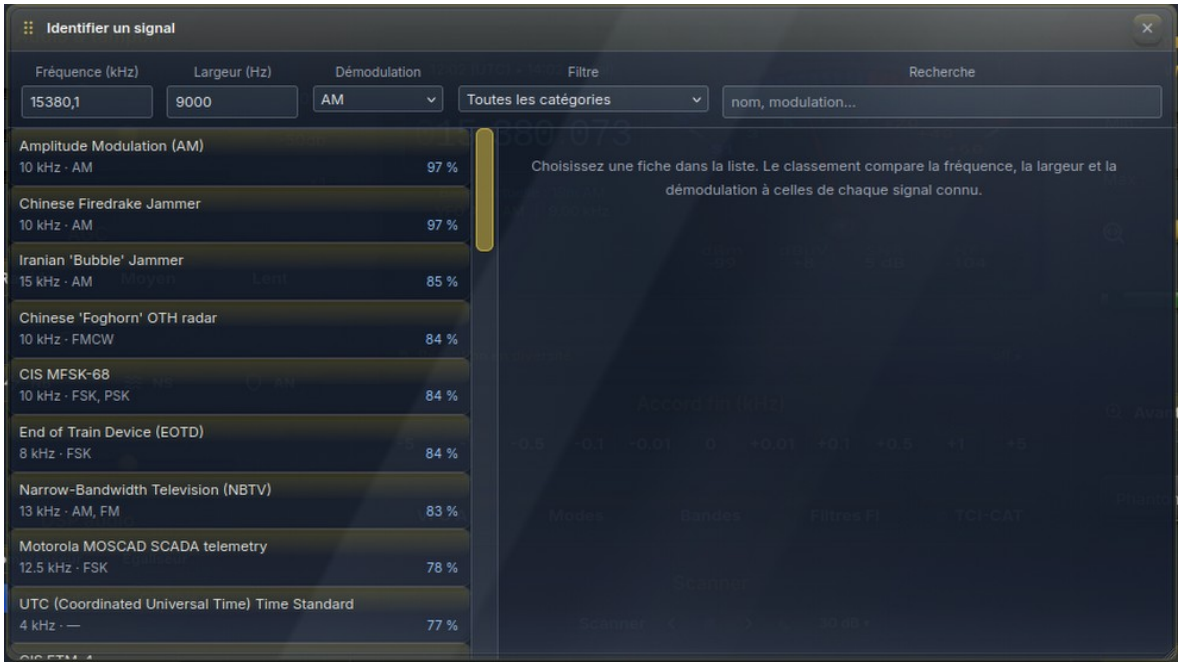


The “Digital signal analysis” module.

- **Automatic analysis: OFF / ON** — the mode is usually recognised in about fifteen seconds.
- **Current measurement** — width, tone spacing, speed, cadence, signal-to-noise ratio.
- **Recognised modes, by relevance** — ▶ opens the matching decoder (with its settings), the **i** icon shows the mode's entry.
- **Recentre & retry** — appears when the signal is too close to the edge of the filter.

12.2 Identify a signal (Artemis database)

The signal database button searches **583 reference signals** for those matching the current frequency, filter width and demodulation, ranked by match percentage. Each entry gives a spectrogram, an **audio sample** to compare, the modulation, the frequency range and a link to the Signal Identification Wiki.



The “Identify a signal” window.

13. Troubleshooting

Symptom	Likely cause	Remedy
No FT8 / FT4 / WSPR decodes	Wrong frequency or band closed	Use **Known frequencies**; check on the waterfall that signals are present
Few FT8 decodes, scattered DT	Time sync	Tick **Auto** next to Sync offset
CW unreadable	Several signals in the filter	Narrow CW filter, a single signal
FAX image slanted	Line sync	⇔ **Auto-align**
FAX image in negative	Polarity	⇅ **Invert**
SSTV: wrong colours or nothing	Wrong sideband or wrong mode	Change sideband; try a forced mode then **Force**
NAVTEX silent	Outside the slot	Wait for the station's slot (broadcast table)
RTTY: random characters	Mark / space swapped	Tick or untick **Invert mark / space**
RTTY: nothing printed	Squelch too high	Lower the **Squelch**
PSK31 / Olivia: broken text	Decoder off-centre	⟳ **Auto-tune Center**
HFDL: no messages	Propagation to the ground station	Try another frequency or another station
RADE: red light	Server service stopped	Tell the ORSAT administrator

Tip : *For any decoder, first check by ear and on the waterfall that a signal is really there and well centred: most failures come from approximate tuning.*